

Optical Breadboard Overview

Thorlabs' honeycomb breadboards are made with the same advanced manufacturing and design methods used for our optical tabletops. We also offer solid aluminum breadboards, which provide a convenient and cost-effective platform for prototyping and optical experiments. Our imperial and metric breadboards range in size from 4" x 6" to 6' x 4' and 100 mm x 150 mm to 2000 mm x 1250 mm, respectively.

Available Options for Thorlabs' Honeycomb Breadboards

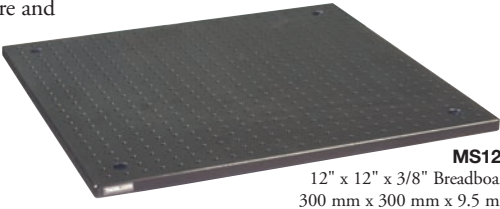
OPTIONAL FEATURES	ULTRALIGHT™ SERIES (Pages 4 - 5)	PERFORMANCE™ SERIES (Pages 6 - 7)	PERFORMANCEPLUS™ SERIES (Pages 8 - 9)
Brushed Stainless Steel Top Plate	N/A	Standard	Standard
Black Finish on Top Plate	Standard	✓	✓
Brushed Steel Finish on Top Plate	N/A	✓	✓
Double-Density Hole Pattern	✓	✓	✓
Blank Top Plate	✓	✓	✓
Access Wells for Standoff Posts	✓	✓	✓
Breadboard Levelers	✓	✓	✓
Nonmagnetic Top and Bottom Plates	Standard	✓	✓
Through Holes/Cutouts	✓	✓	✓
Custom Hole Patterns	✓	✓	✓
Aluminum Honeycomb Core	Standard	✓	✓

Feature Comparison of Honeycomb Breadboards

OPTIONAL FEATURES	ULTRALIGHT™ SERIES (Pages 4 - 5)	PERFORMANCE™ SERIES (Pages 6 - 7)	PERFORMANCEPLUS™ SERIES (Pages 8 - 9)
Thickness	1" or 2.2" (25 mm or 55 mm)	2.4" or 4.3" (60 mm or 110 mm)	2.4" or 4.3" (60 mm or 110 mm)
Flatness	±0.15 mm	±0.10 mm	±0.10 mm
Skin	Aluminum	Magnetic Stainless Steel	Magnetic Stainless Steel
Honeycomb Core	Aluminum	Plated Steel	Plated Steel
Sealed Holes	No	No	Yes
Weight per Unit Area	4.7 lb/ft² or 6.0 lb/ft² (24 kg/m² or 30 kg/m²)	19 lb/ft² or 23 lb/ft² (98 kg/m² or 114 kg/m²)	23 lb/ft² or 28 lb/ft² (114 kg/m² or 137 kg/m²)

Mini-Series Breadboard

Our miniature breadboard features 4-40 (M3 x 0.5) tapped holes on 1/2" (12.5 mm) centers. These solid-aluminum breadboards are complimented by a full range of optomechanical hardware and accessories. These mounting components can be found on pages 74 - 86.



4-40 (M3 x 0.5)
Tapped Holes on
1/2" (12.5 mm) Centers

The HW-KIT5(M) provides over 1,000 assorted 4-40 (M3 x 0.5) 1/4" long setscrews and 1/4", 3/16", 5/16", and 3/8" long socket screws for use with our mini-series optical mounts.

Mini-Breadboard and Hardware Kit

ITEM #	METRIC ITEM #	\$	£	€	RMB	DESCRIPTION
MS12B	MS12B/M	\$ 257.00	£ 185.04	€ 223.59	¥ 2,048.29	12" x 12" High-Density Mini-Breadboard
HW-KIT5	HW-KIT5/M	\$ 92.20	£ 66.38	€ 80.21	¥ 734.83	Hardware Kit for Mini-Series

HW-KIT5
Mini-Series Hardware Kit

Breadboards – 1/2" (12.7 mm) Thick Solid Aluminum

- Custom Sizes and Shapes Available Upon Request
- Matte Black Anodized
- Metric Boards Manufactured and Stocked in Europe
- Flatness: $\pm 0.006"$ (± 0.15 mm) Over any 1 ft² (0.3 m²) Area

Solid Aluminum Construction!

These solid aluminum breadboards provide a convenient platform for assembling prototype optical assemblies, conducting experiments, and mounting small subsystems. They have a black, low-reflective, anodized coating and feature 1/4"-20 (M6 x 1.0) threads on 1" (25 mm) centers.

The boards have a 1/2" (12.5 mm) border and four 1/4" (M6) counterbored clearance holes. The larger boards, which are denoted with double asterisks in the tables below, have a fifth counterbored clearance hole at the center of the board. The plates are machined from a 1/2" (12.7 mm) thick solid aluminum plate for superior flatness.

Because the holes are threaded completely through these breadboards, components can be mounted on both sides. Standoff posts (see pages 20 - 22) allow the breadboards to be mounted above the surface of an optical tabletop (or breadboard) for access to components on both sides.

Stocked for Immediate Shipment

1/4"-20 Taps on 1" Centers, 1/2" Thick*

ITEM #	\$	£	€	RMB	DESCRIPTION
MB4	\$ 49.40	£ 35.57	€ 42.98	¥ 393.72	4" x 6"
MB6	\$ 55.10	£ 39.67	€ 47.94	¥ 439.15	6" x 6"
MB624	\$ 186.00	£ 133.92	€ 161.82	¥ 1,482.42	6" x 24"
MB636	\$ 246.00	£ 177.12	€ 214.02	¥ 1,960.62	6" x 36"
MB648	\$ 306.00	£ 220.32	€ 266.22	¥ 2,438.82	6" x 48"
MB8	\$ 82.00	£ 59.04	€ 71.34	¥ 653.54	8" x 8"
MB810	\$ 143.00	£ 102.96	€ 124.41	¥ 1,139.71	8" x 10"
MB1012	\$ 165.00	£ 118.80	€ 143.55	¥ 1,315.05	10" x 12"
MB12	\$ 161.10	£ 115.99	€ 140.16	¥ 1,283.97	12" x 12"
MB1218	\$ 208.70	£ 150.26	€ 181.57	¥ 1,663.34	12" x 18"
MB1224	\$ 282.30	£ 203.26	€ 245.60	¥ 2,249.93	12" x 24"
MB1236	\$ 505.00	£ 363.60	€ 439.35	¥ 4,024.85	12" x 36"
MB1248	\$ 604.00	£ 434.88	€ 525.48	¥ 4,813.88	12" x 48"
MB18	\$ 315.00	£ 226.80	€ 274.05	¥ 2,510.55	18" x 18"
MB1824	\$ 420.00	£ 302.40	€ 365.40	¥ 3,347.40	18" x 24"
MB1836	\$ 598.00	£ 430.56	€ 520.26	¥ 4,766.06	18" x 36"
MB2424	\$ 537.00	£ 386.64	€ 467.19	¥ 4,279.89	24" x 24"
MB2436	\$ 690.00	£ 496.80	€ 600.30	¥ 5,499.30	24" x 36"
MB2448	\$ 913.00	£ 657.36	€ 794.31	¥ 7,276.61	24" x 48"

*Thickness may vary by $\pm 0.005"$

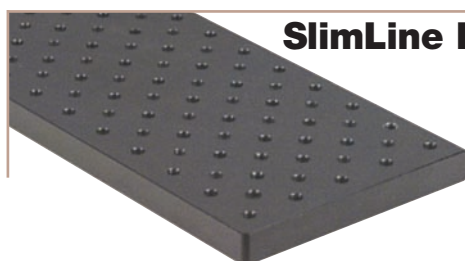
**These boards have a fifth mounting hole in the center

M6 x 1.0 Taps on 25.0 mm Centers, 12.7 mm Thick*

ITEM #	\$	£	€	RMB	DESCRIPTION
MB1015/M	\$ 49.40	£ 35.57	€ 42.98	¥ 393.72	100 mm x 150 mm
MB1515/M	\$ 55.10	£ 39.67	€ 47.94	¥ 439.15	150 mm x 150 mm
MB1560/M	\$ 186.00	£ 133.92	€ 161.82	¥ 1,482.42	150 mm x 600 mm
MB1590/M	\$ 246.00	£ 177.12	€ 214.02	¥ 1,960.62	150 mm x 900 mm
MB15120/M	\$ 306.00	£ 220.32	€ 266.22	¥ 2,438.82	150 mm x 1200 mm
MB2020/M	\$ 82.00	£ 59.04	€ 71.34	¥ 653.54	200 mm x 200 mm
MB2025/M	\$ 143.00	£ 102.96	€ 124.41	¥ 1,139.71	200 mm x 250 mm
MB2530/M	\$ 165.00	£ 118.80	€ 143.55	¥ 1,315.05	250 mm x 300 mm
MB3030/M	\$ 161.10	£ 115.99	€ 140.16	¥ 1,283.97	300 mm x 300 mm
MB3045/M	\$ 208.70	£ 150.26	€ 181.57	¥ 1,663.34	300 mm x 450 mm
MB3060/M	\$ 282.30	£ 203.26	€ 245.60	¥ 2,249.93	300 mm x 600 mm
MB3090/M	\$ 505.00	£ 363.60	€ 439.35	¥ 4,024.85	300 mm x 900 mm
MB30120/M	\$ 604.00	£ 434.88	€ 525.48	¥ 4,813.88	300 mm x 1200 mm
MB4545/M	\$ 315.00	£ 226.80	€ 274.05	¥ 2,510.55	450 mm x 450 mm
MB4560/M	\$ 420.00	£ 302.40	€ 365.40	¥ 3,347.40	450 mm x 600 mm
MB4590/M	\$ 598.00	£ 430.56	€ 520.26	¥ 4,766.06	450 mm x 900 mm
MB6060/M	\$ 537.00	£ 386.64	€ 467.19	¥ 4,279.89	600 mm x 600 mm
MB6090/M	\$ 690.00	£ 496.80	€ 600.30	¥ 5,499.30	600 mm x 900 mm
MB60120/M	\$ 913.00	£ 657.36	€ 794.31	¥ 7,276.61	600 mm x 1200 mm

*Thickness may vary by ± 0.1 mm

**These boards have a fifth mounting hole in the center



SlimLine Breadboards – Double-Density Hole Pattern

The SlimLine series of breadboards provide a double-density 0.5" (12.5 mm) offset hole pattern. This configuration maximizes mounting versatility by doubling the number of tapped holes on the mounting surface. These boards are constructed from a 1/2" (12.7 mm) thick aluminum plate to maintain flatness over the full length. The slim profile allows these boards to fit into small spaces.

Shipped from Stock
in the US and Europe

- Double-Density Hole Pattern
- Custom Sizes Available Upon Request
- Matte Black Anodized Solid Aluminum Construction
- Metric Boards Manufactured and Stocked in Europe

1/4"-20 Taps, 1/2" Offset Hole Pattern, 1/2" Thick*

ITEM #	\$	£	€	RMB	DESCRIPTION
MB424	\$ 161.00	£ 115.92	€ 140.07	¥ 1,283.17	4" x 24"
MB436	\$ 181.00	£ 130.32	€ 157.47	¥ 1,442.57	4" x 36"
MB612	\$ 124.00	£ 89.28	€ 107.88	¥ 988.28	6" x 12"
MB824	\$ 239.00	£ 172.08	€ 207.93	¥ 1,904.83	8" x 24"
MB836	\$ 322.00	£ 231.84	€ 280.14	¥ 2,566.34	8" x 36"

*Thickness may vary by $\pm 0.005"$

M6 x 1.0 Taps, 12.5 mm Offset Hole Pattern, 12.7 mm Thick*

ITEM #	\$	£	€	RMB	DESCRIPTION
MB1060/M	\$ 161.00	£ 115.92	€ 140.07	¥ 1,283.17	100 mm x 600 mm
MB1090/M	\$ 181.00	£ 130.32	€ 157.47	¥ 1,442.57	100 mm x 900 mm
MB1530/M	\$ 124.00	£ 89.28	€ 107.88	¥ 988.28	150 mm x 300 mm
MB2060/M	\$ 239.00	£ 172.08	€ 207.93	¥ 1,904.83	200 mm x 600 mm
MB2090/M	\$ 322.00	£ 231.84	€ 280.14	¥ 2,566.34	200 mm x 900 mm

*Thickness may vary by ± 0.1 mm

Water-Cooled Breadboard

NEW
product



MBC12

- Working Pressure: Rated up to 80 psi
- Working Temperature Range: 0 to 60 °C
- Size: 12" x 12" (300 mm x 300 mm)
- 16' (5 m) of Tubing Included
- Push-to-Fit Tubing Connectors

These breadboards with internal channels for cooling/heating fluid feature a standard 1.0" or 25 mm hole pattern and are ideal for applications involving sensors, detectors, lasers, LEDs, or other devices that may require temperature stabilization. The boards can be used for cooling or heating, depending on requirements. Our water-cooled breadboards are fitted with 1/4" or 6 mm I.D. fittings for imperial and metric boards, respectively. These allow for easy connection to a water supply or chiller unit.

Water-Cooled Breadboard

ITEM #	METRIC ITEM #	\$	£	€	RMB	DESCRIPTION
MBC12	MBC3030/M	\$ 399.00	£ 287.28	€ 347.13	¥ 3,180.03	Water-Cooled Breadboard

UltraLight™ Series of Optical Breadboards (Page 1 of 2)

Specifications

- **Mounting Hole Spacing**
 - Imperial: 1/4"-20 on 1" Centers
 - Metric: M6 on 25 mm Centers
- **Flatness:** ±0.006" (±0.15 mm) Over any 1 ft² (0.30 m²) Area
- **Construction:** Double-Plate, Single Honeycomb-Core, Athermalized Design
- **Outer Plates:** Top 1/4" (6 mm) Thick, Bottom 1/8" (3 mm) Thick
- **Edge of Top Plate to First Hole**
 - Imperial: 1"
 - Metric: 25 mm
- **Maximum Screw Depth:** 5/8" (16 mm) from Breadboard Surface

All Aluminum
Construction



Nonmagnetic UltraLight™ optical breadboards, which offer a high strength-to-weight ratio and excellent thermal stability, are ideal for optical setups where portability and dynamic rigidity are important. They are typically used as lightweight alternatives to

steel breadboards as well as for applications demanding a totally nonmagnetic structure. The options listed here are stocked for immediate shipment. Please visit our website (www.thorlabs.com) to view other size options.

Imperial UltraLight™ Breadboards, 1" Thick

ITEM #	\$	£	€	RMB	DIMENSIONS	WEIGHT*
PBG11102	\$ 511.90	£ 368.57	€ 445.35	¥ 4,079.84	2' x 1' x 1"	11 lbs/5 kg
PBG11118	\$ 618.70	£ 445.46	€ 538.27	¥ 4,931.04	2' x 1.5' x 1"	17 lbs/7.7 kg
PBG11105	\$ 725.60	£ 522.43	€ 631.27	¥ 5,783.03	2' x 2' x 1"	20 lbs/9 kg
PBG11103	\$ 617.60	£ 444.67	€ 537.31	¥ 4,922.27	3' x 1' x 1"	20 lbs/9 kg
PBG11106	\$ 936.20	£ 674.06	€ 814.49	¥ 7,461.51	3' x 2' x 1"	29 lbs/13 kg
PBG11110	\$ 1,095.60	£ 788.83	€ 953.17	¥ 8,731.93	3' x 2.5' x 1"	35 lbs/16 kg
PBG11112	\$ 1,255.60	£ 904.03	€ 1,092.37	¥ 10,007.13	3' x 3' x 1"	42 lbs/19 kg
PBG11107	\$ 1,146.40	£ 825.41	€ 997.37	¥ 9,136.81	4' x 2' x 1"	45 lbs/20.5 kg
PBG11111	\$ 1,357.50	£ 977.40	€ 1,181.03	¥ 10,819.28	4' x 2.5' x 1"	55 lbs/25 kg
PBG11113	\$ 1,569.70	£ 1,130.18	€ 1,365.64	¥ 12,510.51	4' x 3' x 1"	68 lbs/31 kg

*The weights given are for the breadboards alone and do not include packaging.

UltraLight™ Series of Optical Breadboards (Page 2 of 2)



Sizes Shown Here are Stocked
for Immediate Shipment.
Other Sizes are Available up to
4' x 6' (1250 mm x 1800 mm)

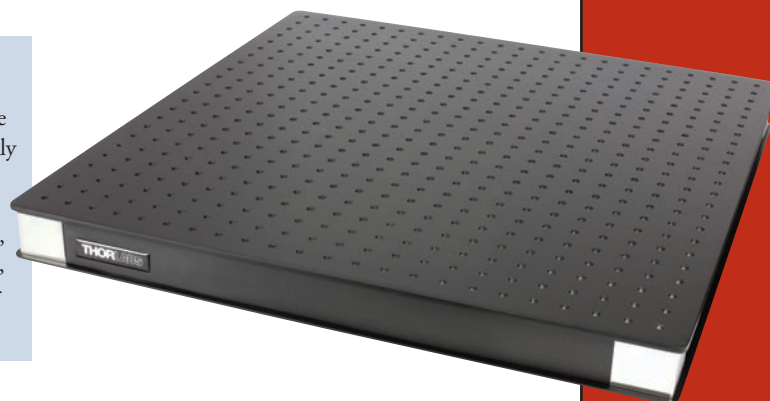
Metric UltraLight™ Breadboards, 25 mm Thick

ITEM #	\$	£	€	RMB	DIMENSIONS	WEIGHT*
PBG51501	\$ 402.80	£ 290.02	€ 350.44	¥ 3,210.32	300 mm x 300 mm x 25 mm	7 lbs/3 kg
PBG51502	\$ 505.30	£ 363.82	€ 439.61	¥ 4,027.24	600 mm x 300 mm x 25 mm	11 lbs/5 kg
PBG51522	\$ 609.20	£ 438.62	€ 530.00	¥ 4,855.32	600 mm x 450 mm x 25 mm	18 lbs/8 kg
PBG51505	\$ 713.10	£ 513.43	€ 620.40	¥ 5,683.41	600 mm x 600 mm x 25 mm	23 lbs/10.5 kg
PBG51503	\$ 607.80	£ 437.62	€ 528.79	¥ 4,844.17	900 mm x 300 mm x 25 mm	31 lbs/14 kg
PBG51506	\$ 917.10	£ 660.31	€ 797.88	¥ 7,309.29	900 mm x 600 mm x 25 mm	33 lbs/15 kg
PBG51510	\$ 1,071.80	£ 771.70	€ 932.47	¥ 8,542.25	900 mm x 750 mm x 25 mm	40 lbs/18 kg
PBG51512	\$ 1,226.40	£ 883.01	€ 1,066.97	¥ 9,774.41	900 mm x 900 mm x 25 mm	51 lbs/23 kg
PBG51507	\$ 1,121.10	£ 807.19	€ 975.36	¥ 8,935.17	1200 mm x 600 mm x 25 mm	37 lbs/17 kg
PBG51511	\$ 1,326.50	£ 955.08	€ 1,154.06	¥ 10,572.21	1200 mm x 750 mm x 25 mm	57 lbs/26 kg
PBG51513	\$ 1,531.90	£ 1,102.97	€ 1,332.75	¥ 12,209.24	1200 mm x 900 mm x 25 mm	66 lbs/30 kg
PBG51514	\$ 1,837.40	£ 1,322.93	€ 1,598.54	¥ 14,644.08	1500 mm x 900 mm x 25 mm	88 lbs/40 kg

*The weights given are for the breadboards alone and do not include packaging.

Additional Size Options Available

The imperial and metric UltraLight breadboards featured on these two pages are stocked for immediate shipment. However, Thorlabs also has the tooling and supplies necessary to manufacture numerous other sizes quickly and efficiently with minimal lead time. Please visit www.thorlabs.com to view our non-standard size options and to place an order. Finally, in addition to these stocked and non-stocked UltraLight breadboard options, Thorlabs can produce truly custom breadboards that offer unique cutouts, hole patterns, and dimensions. Please contact tech support to discuss your application needs.



Imperial UltraLight™ Breadboards, 2.2" Thick

ITEM #	\$	£	€	RMB	DIMENSIONS	WEIGHT*
PBG12118	\$ 654.40	£ 471.17	€ 569.33	¥ 5,215.57	2' x 1.5' x 2.2"	26 lbs/12 kg
PBG12105	\$ 808.60	£ 582.19	€ 703.48	¥ 6,444.54	2' x 2' x 2.2"	27 lbs/12 kg
PBG12106	\$ 1,021.00	£ 735.12	€ 888.27	¥ 8,137.37	3' x 2' x 2.2"	35 lbs/16 kg
PBG12110	\$ 1,180.70	£ 850.10	€ 1,027.21	¥ 9,410.18	3' x 2.5' x 2.2"	44 lbs/20 kg
PBG12111	\$ 1,443.90	£ 1,039.61	€ 1,256.19	¥ 11,507.88	4' x 2.5' x 2.2"	60 lbs/27 kg
PBG12113	\$ 1,656.90	£ 1,192.97	€ 1,441.50	¥ 13,205.49	4' x 3' x 2.2"	71 lbs/32 kg
PBG12114	\$ 1,973.50	£ 1,420.92	€ 1,716.95	¥ 15,728.80	5' x 3' x 2.2"	86 lbs/39 kg

*The weights given are for the breadboards alone and do not include packaging.

Metric UltraLight™ Breadboards, 55 mm Thick

ITEM #	\$	£	€	RMB	DIMENSIONS	WEIGHT*
PBG52502	\$ 540.30	£ 389.02	€ 470.06	¥ 4,306.19	600 mm x 300 mm x 55 mm	13 lbs/6 kg
PBG52522	\$ 644.90	£ 464.33	€ 561.06	¥ 5,139.85	600 mm x 450 mm x 55 mm	18 lbs/8.2 kg
PBG52505	\$ 780.50	£ 561.96	€ 679.04	¥ 6,220.59	600 mm x 600 mm x 55 mm	24 lbs/11 kg
PBG52506	\$ 1,001.40	£ 721.01	€ 871.22	¥ 7,981.16	900 mm x 600 mm x 55 mm	35 lbs/16 kg
PBG52510	\$ 1,156.80	£ 832.90	€ 1,006.42	¥ 9,219.70	900 mm x 750 mm x 55 mm	44 lbs/20 kg
PBG52511	\$ 1,412.90	£ 1,017.29	€ 1,229.22	¥ 11,260.81	1200 mm x 750 mm x 55 mm	55 lbs/25 kg
PBG52513	\$ 1,619.00	£ 1,165.68	€ 1,408.53	¥ 12,903.43	1200 mm x 900 mm x 55 mm	57 lbs/27 kg
PBG52508	\$ 1,413.00	£ 1,017.36	€ 1,229.31	¥ 11,261.61	1500 mm x 600 mm x 55 mm	60 lbs/27 kg
PBG52514	\$ 1,925.90	£ 1,386.65	€ 1,675.53	¥ 15,349.42	1500 mm x 900 mm x 55 mm	66 lbs/30 kg
PBG52521	\$ 3,182.10	£ 2,291.11	€ 2,768.43	¥ 25,361.34	1800 mm x 1250 mm x 55 mm	145 lbs/66 kg

*The weights given are for the breadboards alone and do not include packaging.